

YUYANG (ROY) WANG

AI APPLICATION ENGINEER – BRIDGING ML, FULL-STACK, AND HUMAN-CENTERED DESIGN

jg971402@gmail.com
 github.com/RoyWG925
 Portfolio: nucleus-io.vercel.app

EDUCATION

National Taiwan Normal University (NTNU)
SEP 2021 - JUN 2026

Bachelor of Science in Learning Sciences

GPA: 3.47 / 4.3 (Overall) | 4.12 / 4.3 (Senior Fall 2024 — Ranked 4th in class) | Last 2 Years: ~3.78 / 4.3

Relevant: Deep Learning, Generative AI Application Engineering, Discrete Mathematics

PROJECTS & RESEARCH

Event-driven Sentiment & Market Performance
JAN 2025 - DEC 2025

NLP RESEARCH · NTNU

- Designed a **SQLite data pipeline** to crawl, clean, and store **20,000+ posts** from PTT Stock Board
- Fine-tuned a **BERT-based sentiment classifier** on domain-specific financial text, improving **F1 from 0.23 → 0.68** through class-imbalance handling and custom tokenization
- Conducted **time series analysis** correlating sentiment volatility with TAIEX fluctuations for event-driven trading research

Semantic Segmentation Model Comparison
JAN 2024 - MAY 2024

COMPUTER VISION RESEARCH · NTNU

- Trained **SegFormer** and **MobileNet-UNet** on **BDD100K** (10K annotated driving images)
- Systematically evaluated with **confusion matrices, mIoU, per-class accuracy**, identifying SegFormer's edge on small-object classes

Chewsy — Real-time Multiplayer Restaurant Decision App
2025

REACT · FIREBASE · FRAMER MOTION

- Built a Tinder-style swiping app for groups to find restaurant matches, with **live Firebase state sync** across users
- Implemented QR-based room joining, anonymous auth, and spin-wheel tiebreaker with smooth animations

NUCLEUS.IO — Interactive 3D Portfolio
2025

THREE.JS · WEBGL · CUSTOM SHADERS

- Engineered a gamified portfolio: fly a spaceship through space, collide with neon text, collect project orbs
- Built custom post-processing (Bloom), physics-based text collision, and responsive UI overlays

GigaJoule: Scrap Matrix (Engine)
2025 - PRESENT

GODOT ENGINE · GDSCRIPT

- Designing emergent gameplay with multi-cell item synergy, dynamic grid adjacency, and real-time settlement logic

TECHNICAL SKILLS

AI / ML	PyTorch, Transformers (BERT, SegFormer), NLP, Sentiment Analysis, Computer Vision, CUDA
PROGRAMMING	Python, JavaScript / TypeScript, GDScript, SQL
WEB / FULL-STACK	React, Tailwind CSS, Firebase, Vite, Three.js, WebGL, Canvas API
DATA ENGINEER	Pandas, NumPy, SQLite, Web Scraping (BeautifulSoup)
TOOLCHAIN	Git / GitHub, Docker, Vercel, Cursor (AI-augmented dev)

LANGUAGES

Chinese — Native

English — Intermediate (TOEIC 720)

WHAT SETS ME APART

I combine **NLP + Computer Vision research** with **full-stack product building** and a **Learning Sciences background** in human cognition. I don't just build models — I build products people can use. This intersection of AI engineering, creative coding, and understanding how humans learn makes me effective in AI application, EdTech, and HCI-adjacent roles.